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Project No. 项目编号	TXW550039N1-CTP
Customer 客户名称	
Module No. 客户型号	
Product type 产品内容	Standard LCD Module TFT: 720*RGBx1280Dots 5.5”TFT LCD + CTP

客户确认Customer Approval	
项目负责人Project Manager	
品质主管Director of Quality	
采购工程师Purchasing Engineer	

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1. Introduction

1.1 Scope of application

LCD specification: Dots 720xRGBx1280.
As to basic specification of the driver IC, refer to the IC (Sitronix:ST7703) specification and data book.
All material & processing of the LCD module should be Lead Free.

1.2 TFT features:

Structure: TFT PANNEL+IC +FPC+BL +CTP;
IPS Type LCD
720dot-segment and 1280 dot-common outputs;
16.7M Color can be selected by software;
White LED back light;
4-lane MIPI interface;

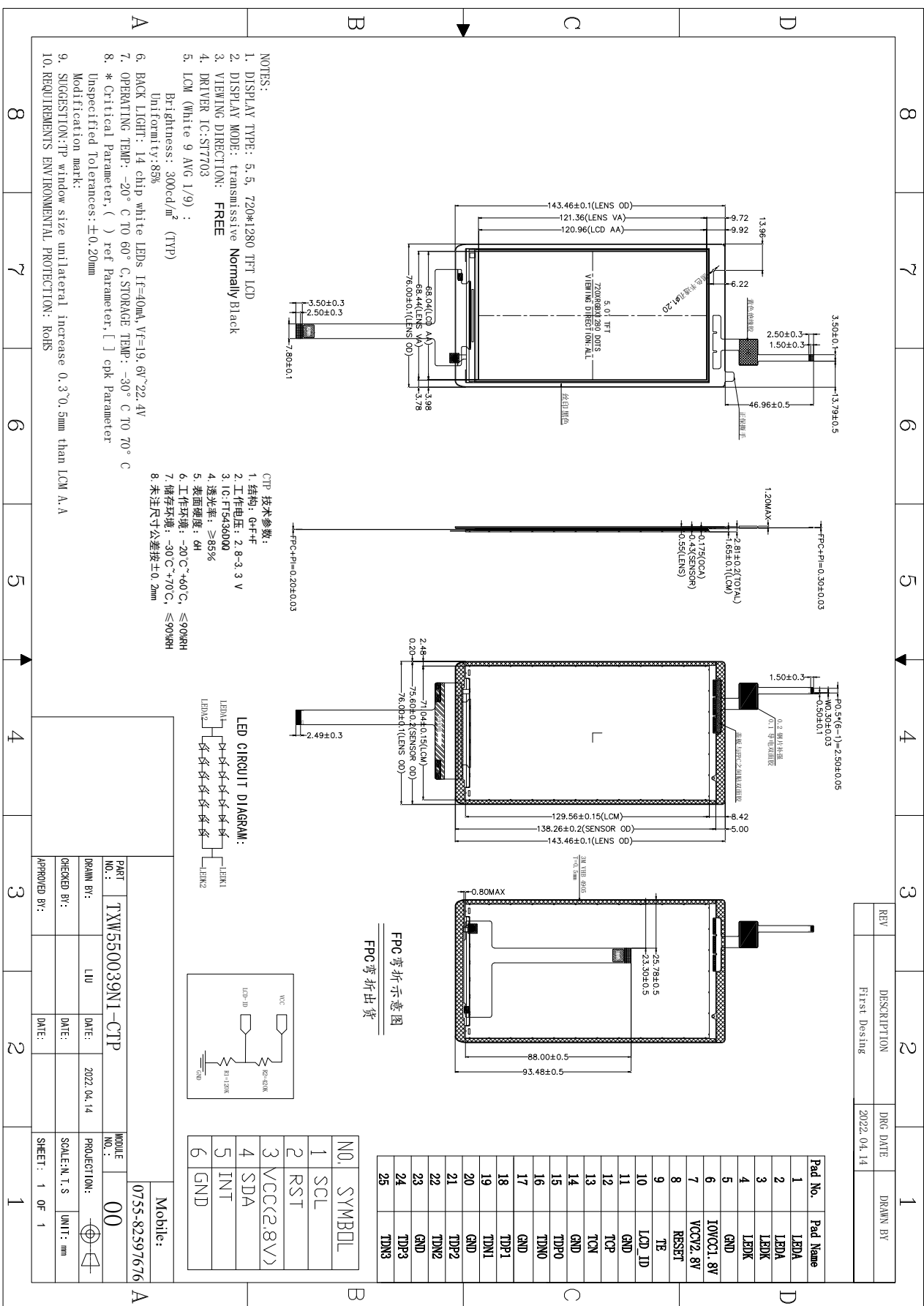
1.3 Applications:

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2. LCM General specification

ITEM	Standard value	Unit
LCD Type	Normally black	--
Drive element	TFT active matrix	--
Number of pixels	720*3RGB(H)X1280(V)	Dots
Pixel arrangement	RGB stripe	--
pixel size	94.5 x 94.5	um
Active area	68.04(W) × 120.96(H)	mm
Viewing direction	ALL O'CLOCK	-
TFT Driver IC	ST7703	
TFT interface	4-lane MIPI Interface	-
Approx. Weight	TBD	g
Module Size(W*H*T)	71.04(W) ×129.56(H) ×1.65(T)	mm
LCM+CTP Size(W*H*T)	76.00(W) ×149.00(H) ×2.81(T)	mm
Touch structure	G+F+F	
Touch Driver IC	FT5436DQQ	-
Touch Interface	I2C	

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3.Absolute Maximum Rating

Characteristics	Symbol	Min.	Max.	Unit
LCM Operating Temperature	T _{OPR}	-20	+60	°C
LCM Storage Temperature	T _{STG}	-30	+70	°C
TP Operating Temperature & Humidity(20% ~ 90%RH)	T _{OPR}	-20	+60	°C
TP SStorage Temperature & Humidity(20% ~ 90%RH)	T _{STG}	-30	+70	°C
Humidity	RH	-	90	%

4.Electrical Characteristics

4.1 TFT DC Characteristics

Characteristics	Symbol	Min.	Typ.	Max.	Unit
Supply Voltage for I/O	VDDIO	1.65	1.8	2.0	V
Supply Voltage for(DC/DC)	VDD	2.5	2.8	3.3	V
Supply Voltage for(DC/DC)	AVDD				V
Supply Voltage for(DC/DC)	AVEE				V
Current Consumption	I _{DD}	-	-	-	mA
	I _{DD-SLEEP}	-	-	-	uA

4.2 Back-Light Unit Characeristics

The back-light system is an edge-lighting type with white 14 LEDs. The characteristics of the back-light are shown in the following tables.

Characteristics	Symbol	Min.	Type	Max.	Unit	Notes
Forward Voltage	V _F	19.6	--	22.4	V	-
Forward current	I _F	--	40	-	mA	-
Luminance(With LCD+CTP)	L _v		300	--	cd/m ²	-
LED life time	N/A	----	30,000	--	Hr	Note 1

Note:

- (1) The “LED life time” is defined as the module brightness decrease to 50% of original brightness at I_L=20mA/LED. The LED life time could be decreased if operating I_L is larger than 25mA/LED.

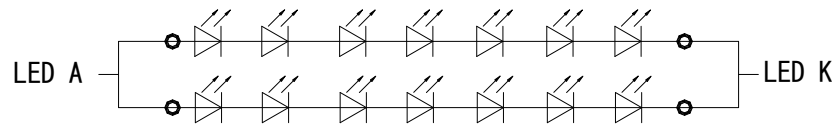
Backlight circuit diagram shown in below:

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5. Module Function Description

Pin No.	Symbol	LCM Functional	Notes
1-2	LEDA	Power supply for backlight anode input terminal.	
3-4	LEDK	Power supply for backlight cathode input terminal.	
5	GND	Power Ground	
6	IOVCC 1.8V	I/O and interface power supply (IOVCC): 1.65V to 2.0V	
7	VCI 2.8V	Analog power supply (VCI): 2.5V to 3.3V	
8	RESET	Reset signal input terminal. Active at 'L'.	
9	TE	Frame head pulse for tearing effect.	
10	LCD-ID	ID read PIN for ID circuit.	
11	GND	Power Ground	
12	CLKP	High speed interface clock differential signal input pins	
13	CLKN	High speed interface clock differential signal input pins	
14	GND	Power Ground	
15	D0P	High speed interface data differential signal input pins	
16	D0N	High speed interface data differential signal input pins	
17	GND	Power Ground	
18	D1P	High speed interface data differential signal input pins	
19	D1N	High speed interface data differential signal input pins	
20	GND	Power Ground	
21	D2P	High speed interface data differential signal input pins	
22	D2N	High speed interface data differential signal input pins	
23	GND	Power Ground	
24	D3P	High speed interface data differential signal input pins	
25	D3N	High speed interface data differential signal input pins	

Pin No.	Symbol	TP Functional	Notes
1	CTP_SCL	CTP_SCL_PIN	
2	CTP_RESET	CTP_RESET_PIN	
3	CTP_VDD 2.8V	CTP_INT_PIN	
4	CTP_SDA	CTP_SDA_PIN	
5	CTP_INT	CTP_INT_PIN	
6	GND	Power Ground	

6. Timing Characteristics

If RESX line is held low (and stable) by the host during power on, then the RESX must be held low for minimum 10 μ sec after both VDD1, VDD2 and VDD3 have been applied.

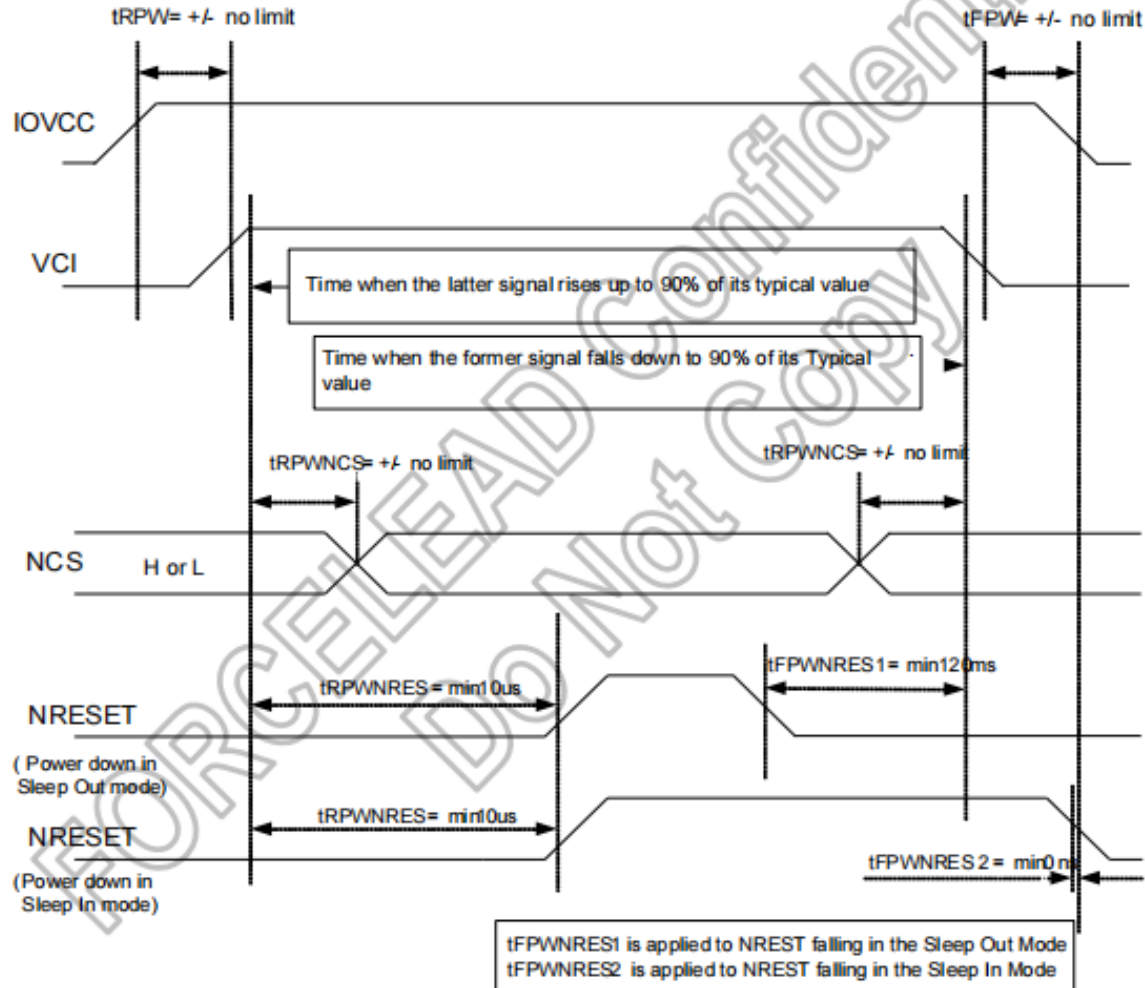
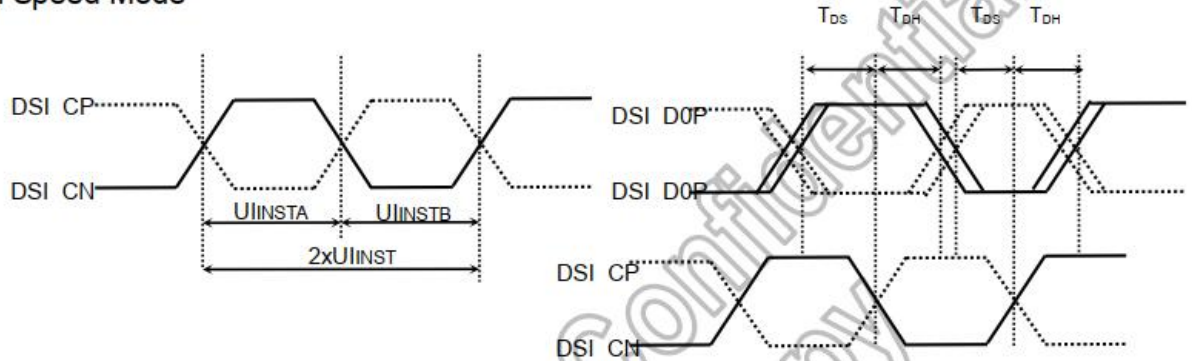


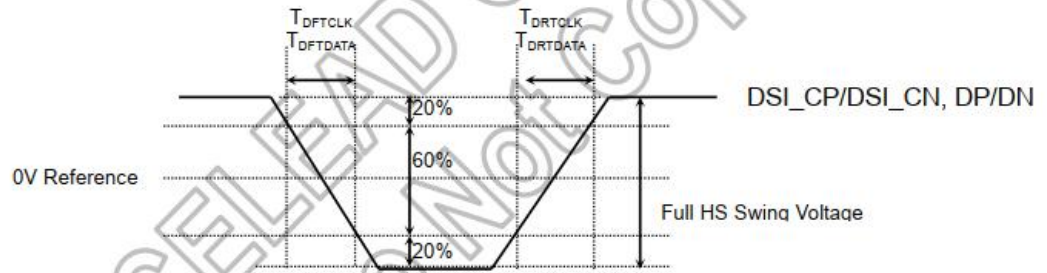
Figure 5.34: Case 2: RESX line is held low by host at power on

LCM

High Speed Mode



DSI clock timing Characteristics



Rising and falling time on clock and data channel

(VSSA=0V, IOVCC=1.65V to 3.3V, VCI=2.5V to 3.3V, $T_A = -30$ to 70°C)

Signal	Item	Symbol	Spec.			Unit
			Min.	Typ.	Max.	
DSI_CP/ DSI_CN	Double UI instantaneous	$2xU_{INST}$	TBD	-	25	ns
	UI instantaneous	U_{INSTA} U_{INSTB}	TBD	-	12.5	ns
DP/DN	Data to clock setup time	T_{DS}	$0.15xUI$	-	-	ps
	Data to clock hold time	T_{DH}	$0.15xUI$	-	-	ps
DSI_CP/ DSI_CN	Differential rise time for clock	T_{DRTCLK}	150	-	$0.3UI$	ps
	Differential fall time for clock	T_{DFTCLK}	150	-	$0.3UI$	ps
DP/DN	Differential rise time for data	$T_{DRTDATA}$	150	-	$0.3UI$	ps
	Differential fall time for data	$T_{DFTDATA}$	150	-	$0.3UI$	ps

Reset input timing

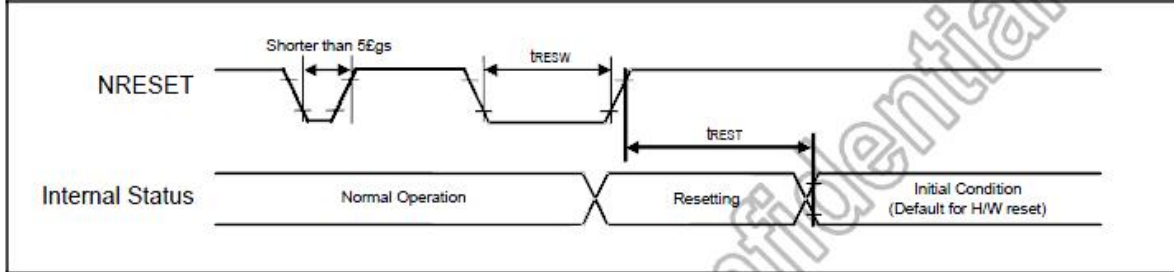


Figure 7.8: Reset input timing

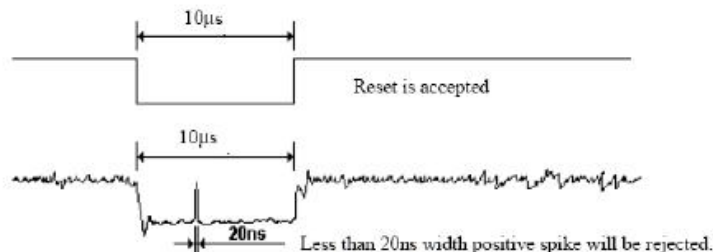
Symbol	Parameter	Related Pins	Spec.			Note	Unit
			Min.	Typ.	Max.		
tRESW	Reset low pulse width ⁽¹⁾	NRESET	10	-	-	-	μs
tREST	Reset complete time ⁽²⁾	-	15	-	-	When reset applied during SLPIN mode	ms
		-	120	-	-	When reset applied during SLPOUT mode	ms

Table 7.8: Reset Input Timing

Note: (1) Spike due to an electrostatic discharge on NRESET line does not cause irregular system reset according to the following table.

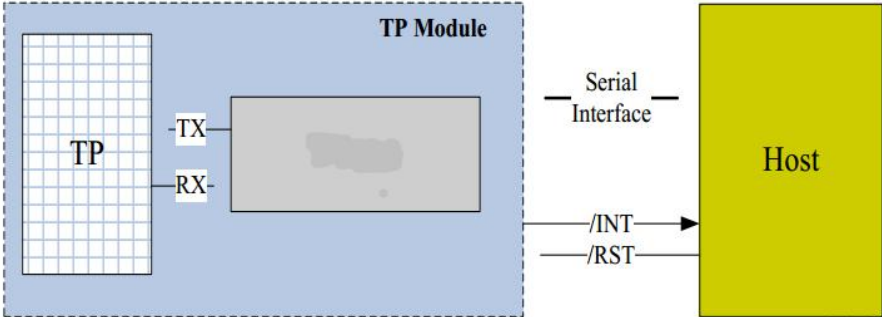
NRESET Pulse	Action
Shorter than 5 μs	Reset Rejected
Longer than 10 μs	Reset
Between 5 μs and 10 μs	Reset Start

- (2) During the resetting period, the display will be blanked (The display is entering blanking sequence, which Maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode) and then return to Default condition for H/W reset.
- (3) During Reset Complete Time, ID and VCOM value in OTP will be latched to internal register during this period. This loading is done every time when there is H/W reset complete time (tREST) within 15ms after a rising edge of NRESET.
- (4) Spike Rejection also applies during a valid reset pulse as shown as below:

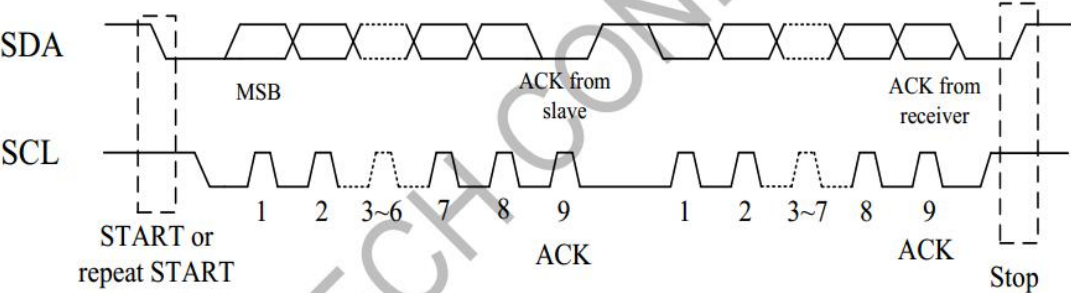


- (5) It is necessary to wait 15msec after releasing NRESET before sending commands. Also Sleep Out command cannot be sent for 120msec.

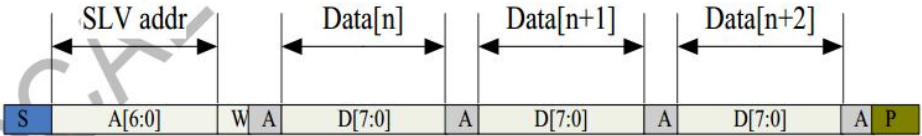
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Host Interface Diagram



I2C Serial Data Transfer Format



I2C master read, slave write

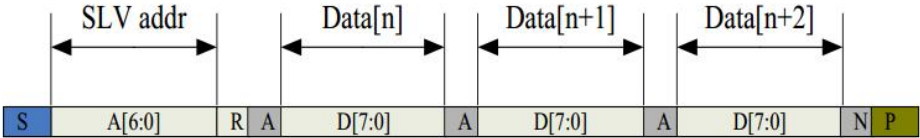


Figure 2-6 I2C master read, slave write

lists the meanings of the mnemonics used in the above figures.

I2C Interface Timing Characteristics is shown in Table 2-2.

Table 2-2 I2C Timing Characteristics

Parameter	Min	Max	Unit
SCL frequency	0	400	KHz
Bus free time between a STOP and START condition	1.3		us
Hold time (repeated) START condition	0.6		us
Data setup time	100		ns
Setup time for a repeated START condition	0.6		us
Setup Time for STOP condition	0.6		us

7.Optical Characteristics

Item		Symbol	Conditions	Specifications			Unit	Note
				Min.	Typ.	Max.		
Transmittance (without DBEF)		T%	Viewing normal angle $x = y = 0$		3.5	--	%	All left side data are based on INX's following condition (at 25 °C) 1.LC : AAS 2.Light Source : INX BLU Spectrum. 3.CF / TFT side Film : SRW062APN1HC5 / SRW062APN1 4.Machine : DMS 803 (Cono Scope for View Angle) 5. VLC white > 4.5 V VLC dark < 0.2 V
Contrast Ratio		CR		600	1000	--	--	
Response Time		T _{on} +T _{off}		--	25	35	ms	
Viewing Angle	Hor.	x+	CR ≥10 at 25 °C	75	80	--	deg.	
		x-		75	80	--		
	Ver.	y+		75	80	--		
		y-		75	80	--		
CF Only Color Chromaticity (CIE1931)	Red	X _R	Viewing normal angle $x = y = 0$ at 25 °C	0.635	0.655	0.675	--	
		Y _R		0.301	0.321	0.341		
	Green	X _G		0.237	0.257	0.277		
		Y _G		0.539	0.559	0.579		
	Blue	X _B		0.119	0.139	0.159		
		Y _B		0.071	0.091	0.111		
	White	X _W		0.274	0.294	0.314		
		Y _W		0.294	0.314	0.334		
Color Gamut	CG			60	68	--	%	

*Note

(1)Definition of Contrast Ratio (CR):

The contrast ratio can be calculated by the following expression.

Contrast Ratio (CR) = L63 / L0

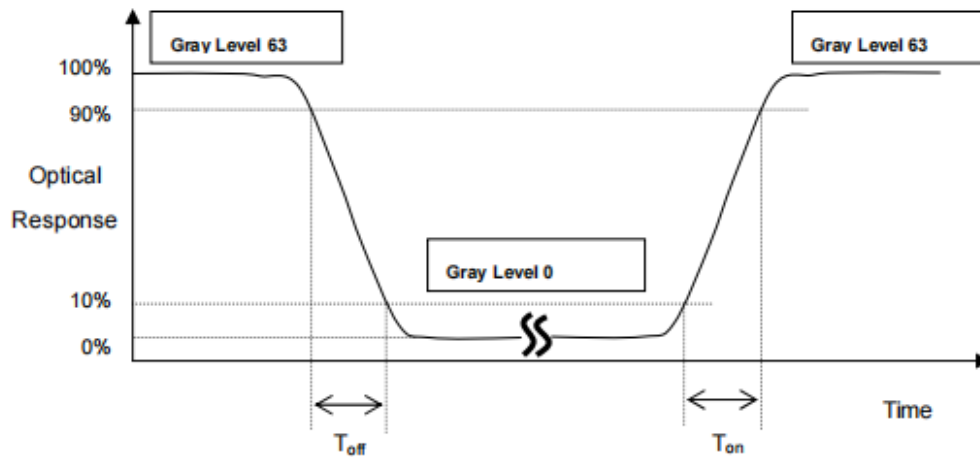
L63: Luminance of gray level 63

L 0: Luminance of gray level 0

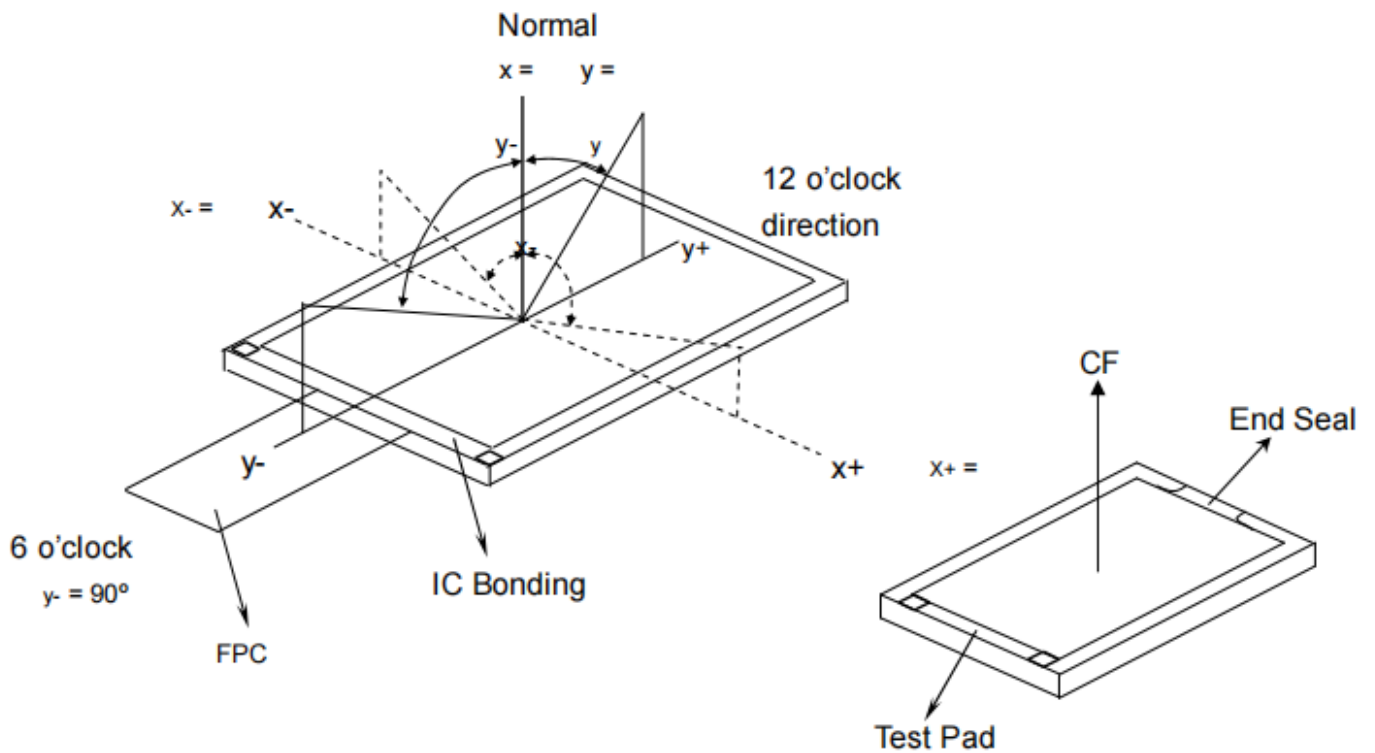
CR = CR (5)

CR (X) is corresponding to the Contrast Ratio of the point X at Figure in Note (5).

*Note (2) Definition of Response Time (T_{on} , T_{off}):

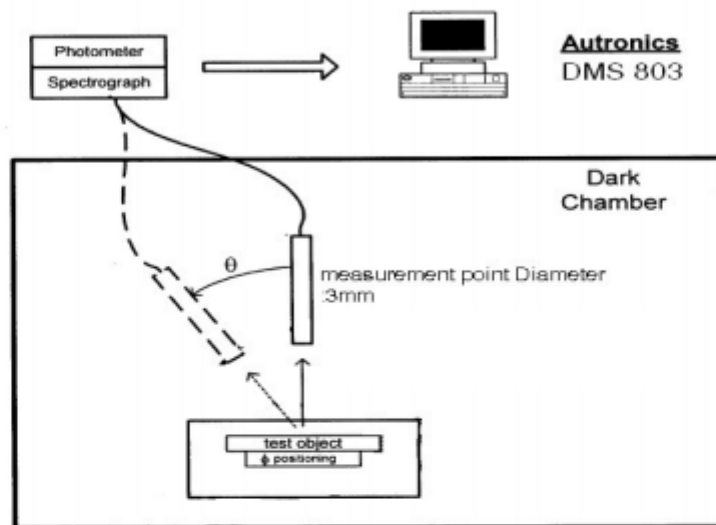


*Note(3) Definition of Viewing Angle

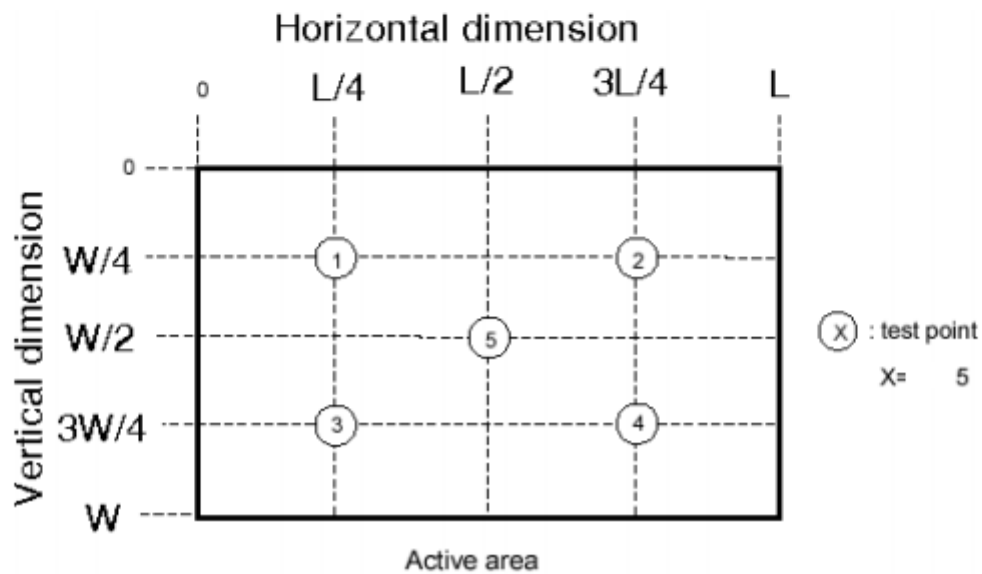


*Note (4) Measurement Set-Up:

The LCD module should be stabilized at a given temperature for 20 minutes to avoid abrupt temperature change during measuring. In order to stabilize the luminance, the measurement should be executed after lighting Backlight for 20 minutes in a windless room.



*Note (5)



8. Reliability Test Item

No.	Test Item	Test Condition	Notes
1	High Temp. Storage	+70℃ / 96H	1. Functional test isOK. Missing Segment,short, unclear segment non-display,display abnormally and liquid crystal leakare un-allowed. 2. No low temperature bubbles,end seal loose andfall, frame rainbow.
2	Low Temp. Storage	-30℃ / 96H	
3	High Tempe. Operating	+60℃ / 96H	
4	Low Tempe. Operating	-20℃ / 96H	
5	High Temperature /Humidity storage	50℃ x 90%RH /96H	
6	Thermal and cold shock	Static state, -20℃（30min）~60℃ （30min）, 50 cycles	

9. Packing Method----TBD

- END -